An applied pull is not always the net force

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Driving force / ES: fuerza impulsora / PT: força motriz

The pull before resistance is subtracted.

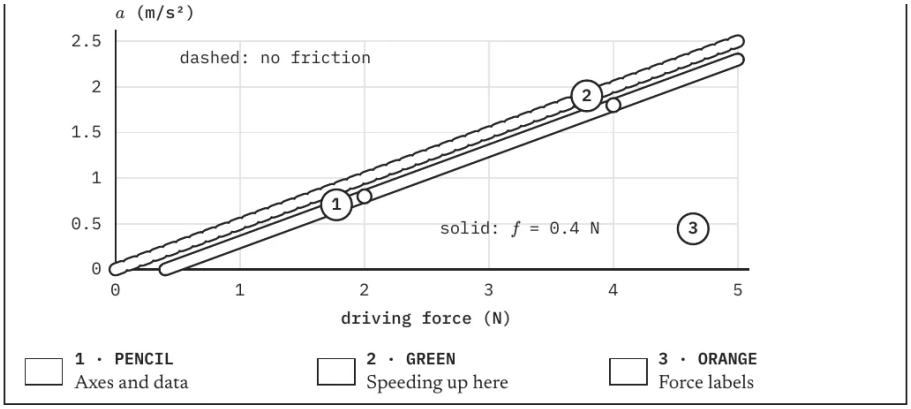
Also on your note page: Net force, Slope, Friction offset.

B READ THE RELATIONSHIP

Fnet = Fdrive − f; a = Fnet/M

As you read, find: driving pull · friction · net force.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg system slides with 0.4 N of opposing friction. Compare acceleration with driving forces of 2 N and 4 N .

Given M = 2 kg f = 0.4 N F drive = 2 N , then 4 N

Rule Subtract friction. Then a = F net / M .

Work a 1 = (2 − 0.4)/2 = 0.8 a 2 = (4 − 0.4)/2 = 1.8

Answer 0.8; 1.8 m/s² Driving force doubles; a does not.

E SAVE FOR THE FINAL MEETING

A proposed push on XJ must be added to every other external force. Only the net force belongs in the calculation of acceleration.

Read the physics: Driving force

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Keep the total moving mass the same. With a cart and hanger, move mass from the cart to the hanger. This changes the driving pull while keeping the moving mass constant.

02 HOW IT WORKS

The driving force is one pull. Subtract any opposing friction to find the net force. Divide net force by total mass to find acceleration.

03 WHAT TO CHECK

At fixed mass, doubling the net force doubles acceleration. Doubling the driving pull need not double acceleration when friction is present.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg system slides with 0.4 N of opposing friction. Compare acceleration with driving forces of 2 N and 4 N . 0.8; 1.8 m/s² Driving force doubles; a does not.

Model check: Fnet = Fdrive − f; a = Fnet/M.

05 CONNECT TO XJ

A proposed push on XJ must be added to every other external force. Only the net force belongs in the calculation of acceleration.

SAY IT BEFORE YOU WRITE

Which force goes in Newton's second law? The driving pull is ___. After friction, the net force is ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Driving force

ES: fuerza motriz / PT: força motriz

Meaning in my words	How can you tell?
How would you explain driving force to a friend? In my words, this means ___.	A rope pulls right while friction acts left. Which one is the driving force on the cart? I can tell because ___.
Example or sketch	**Non-example and why**
Draw a 2 N driving pull and 0.4 N of friction. Which arrow is the driving force? My example shows ___.	Is the hanging weight always the same as the rope tension on the cart? I would change ___ because ___.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words	How can you tell?
How would you explain net force to a friend? In my words, this means ___.	Two people push one cart in opposite directions. Why do the arrow directions matter? I can tell because ___.
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both? My example shows ___.	A student adds those pushes to get 10 N right. Which push did they count the wrong way? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Slope

ES: pendiente / PT: inclinação

Meaning in my words	How can you tell?
How would you explain slope to a friend? In my words, this means ___.	Which cart has less mass if its acceleration rises more per newton? I can tell because ___.
Example or sketch	**Non-example and why**
Net force rises 4 N and acceleration rises 2 m/s². What mass fits that slope? My example shows ___.	A partner reads the acceleration at one point and calls it slope. What two changes do you need? I would change ___ because ___.

Friction offset

ES: desfase por fricción / PT: deslocamento devido ao atrito

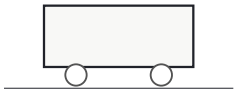
Meaning in my words	How can you tell?
How would you explain friction offset to a friend? In my words, this means ___.	If the pull is 3 N and opposing friction is 1 N, how much force accelerates the cart? I can tell because ___.
Example or sketch	**Non-example and why**
For a sliding 2 kg cart with 1 N opposing friction, what acceleration would a 3 N pull produce? My example shows ___.	A student uses the full 3 N pull as the net force. What did they leave out? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the pull and opposing friction. Label driving force on one arrow and net force on a separate arrow.

Labels you can use: driving pull · friction · net force.



Tell a partner

The driving pull is ___. After friction, the net force is ___.

Which run changes more?

12

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when net forces differ.

CHOOSE ONE JOB TOGETHER

FLIGHT-TEST ENGINEER ☐

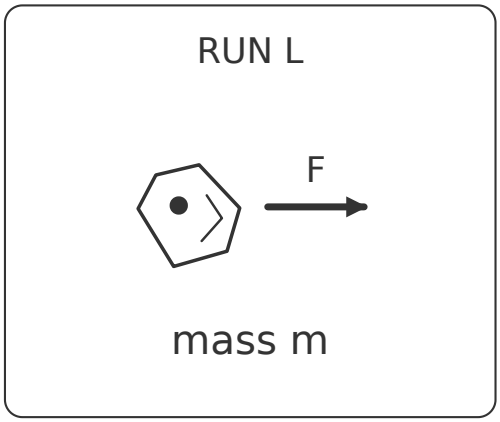
Add velocity-change arrows for both runs. Recommend the larger change and name what stayed the same.

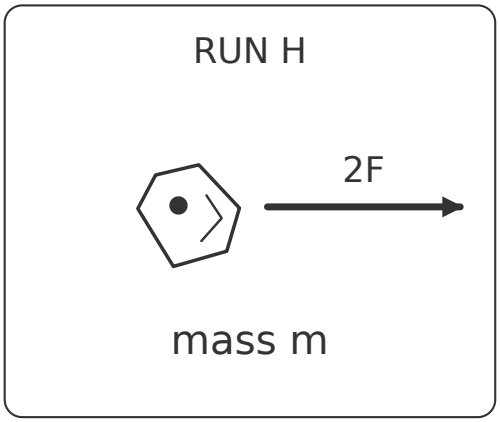
TEST-RANGE BROADCASTER ☐

Tell viewers which run changes velocity more. Ask whether doubling motor pull always doubles the net force.

Word help: net force / drag / velocity change

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.





01 OUR FIRST CALL

How do the two velocity changes compare? Point to the force arrows.

The velocity change is ___ if ___ stays the same.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Do the new readings still support your comparison? Explain what you must check.

Now we would say ___ because ___.

Switch speakers. Explain the difference between doubling one applied force and doubling the net force.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks